

[Lab] PyTorch on multiple versions of Python

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Due: Before the end of today's lab session

Evaluation:

- Show your Python code & results to the Professor.
- Answer his questions on: the theory (seen in lecture), the code, and the results.

Remark:

- Lab tasks must be performed only in groups of two or three (preferably of three). Groups of one or more than three are forbidden.
 - No plagiarism. If plagiarism happens, both the “lender” and the “borrower” will have a zero.
 - Code yourself from scratch **following the theory explained in class**.
 - Study the theory.
 - There is NO make-up session.
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1 Multiple versions of Python

1.1 Tasks

1. Download and install Python version 3.12.10 and Python version 3.7.9 following the instructions given in **python_installation_venv_pip.pdf**.
2. Install *virtualenv* for Python virtual environment.
3. Create a Python virtual environment for each of the two above Python versions.

2 PyTorch

Note: Make sure that you activated one of the two Python virtual environments before you start to work on the following tasks.

2.1 Tasks

1. Follow the instructions given in **PyTorch.pdf** file.
2. Among all the results that you obtain from these instructions, show the following results to Professor **for both Python virtual environments**:
 - The results of Section 3 (Linear regression),
 - The results of Section 4 (Classification regression),
 - The results of Section 5 (Fine tuning deep learning models with PyTorch),
 - The results of Section 8 (Implementing a CNN).